

TERRA-DRIP

Modular Smart Irrigator — Source Code Excerpt

This excerpt shows the core watering-decision logic from the Arduino sketch controlling TerraDrip. The full source (150+ lines, including OLED display handling, safety lockout, and watchdog recovery) is available via the QR code at the bottom of this page.

```
int readMoisturePercent() {
    long total = 0;
    const int samples = 5;
    for (int i = 0; i < samples; i++) {
        total += analogRead(SOIL_PIN);
        delay(5);
    }
    int raw = total / samples;
    int percent = map(raw, DRY_VALUE,
                     WET_VALUE, 0, 100);
    percent = constrain(percent, 0, 100);
    return percent;
}

void loop() {
    wdt_reset();
    unsigned long now = millis();

    if (pumpState && (now - pumpStartTime
                     >= PUMP_RUN_TIME)) {
        pumpState = false;
        pumpOffTime = now;
        digitalWrite(PUMP_PIN, HIGH);
    }

    if (now - lastCheck >= CHECK_INTERVAL) {
        lastCheck = now;
    }
}
```

```
int moisture = readMoisturePercent();

bool soakOver = (now - pumpOffTime
                 >= SOAK_TIME);

if (!pumpState && !lockedOut &&
    soakOver &&
    moisture < MOISTURE_ON_THRESHOLD) {
    pumpState = true;
    pumpStartTime = now;
    digitalWrite(PUMP_PIN, LOW);
    consecutiveCycles++;

    if (consecutiveCycles >=
        MAX_CONSECUTIVE_CYCLES) {
        lockedOut = true;
        pumpState = false;
        digitalWrite(PUMP_PIN, HIGH);
    }
}

if (moisture >= MOISTURE_ON_THRESHOLD) {
    consecutiveCycles = 0;
    lockedOut = false;
}
}
```



Scan for full source code

Complete 150+ line sketch, syntax-highlighted,
with project explanation.

draven-ashcroft.github.io/Terra-Drip/